

Mathematical Scoping for Art Direction

Redefining the Variable V_w (Viewport Width) in the CSS Evaluation Context

1. The Functional Remapping Equation

In standard HTML/CSS, a media query is a boolean function $Q(V_w)$ evaluated against the global window object:

$$Q_{\text{global}}(V_w) = \begin{cases} \text{True,} & \text{if } V_w \leq \text{Threshold} \\ \text{False,} & \text{otherwise} \end{cases}$$

When an `<svg>` is loaded as a protected asset via an `<img>` tag, the internal CSS parser does **not** inherit the global window V_w . The layout engine executes a substitution mapping the container's rendered width (C_w) as the new independent variable:

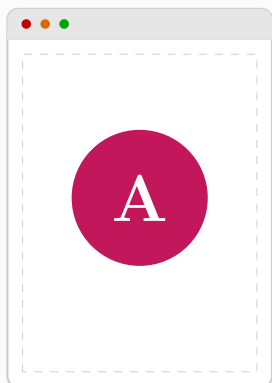
$$V_{w(\text{svg})} := C_w(\text{Host } \text{ Node}) \Rightarrow Q_{\text{local}}(V_{w(\text{svg})})$$

2. Practical Visual Proof: Single CDN Asset, Multiple Contexts

Context A: Sidebar Menu

Host Container: $C_w = 60\text{px}$
Eval: $60\text{px} \leq 100\text{px}$ (TRUE)

$C_w = 60\text{px}$



Internal Engine triggers:
@media (max-width: 100px)
Hides all text elements, scales and centers the monogram for tight spaces.

Context B: Article Body

Host Container: $C_w = 300\text{px}$
Eval: $100\text{px} < 300\text{px} \leq 400\text{px}$ (TRUE)

$C_w = 300\text{px}$



Internal Engine triggers:
@media (max-width: 400px)
Reveals company name, switches to horizontal flex-layout spacing.

Context C: Hero Header

Host Container: $C_w = 800\text{px}$
Eval: $800\text{px} \leq 400\text{px}$ (FALSE)

$C_w = 800\text{px}$



Internal Engine triggers:
Default root styles
Shows full stacked lockup with tagline for maximum brand visibility.

■ Conclusion: The exact same static URL resolves to 3 dynamic layout states driven purely by local mathematics.